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Cascading failure resilience analysis and recovery of automotive manufacturing supply chain networks considering enterprise roles

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ABSTRACT

With the rapid development of global trade, the member enterprises in the automotive manufacturing supply chain are widely spread around the world. As a result, the scale of the automotive manufacturing supply chain network has become increasingly large, and the structure has become more complex. At the same time, the connection between enterprises is becoming closer. On one hand, the close business relationships between enterprises and the vast network structure accelerate the operation efficiency of the automotive manufacturing supply chain network. On the other hand, it also exposes the entire network system to high risks of cascading failures when facing disruptions due to enterprise production interruptions. Therefore, this work aims to establish a realistic cascading failure model for the automotive manufacturing supply chain network to explore its cascading failure characteristics. Based on this model, a two-stage recovery strategy is proposed to enhance the network's cascading failure resilience. Experimental results demonstrate the following findings: (1) underload nodes that emerge after load redistribution are the main cause of cascading failure resilience degradation in automotive manufacturing supply chain networks; (2) cascading failure resilience is negatively correlated with the load-exponential coefficient, the capacity lower-limit coefficient, and the attenuation-tolerance coefficient; (3) among four different types of attacks, cascading failures caused by the high eigenvector centrality attack strategy and the high degree attack strategy impose the most severe damage on the automotive manufacturing supply chain network; (4) the proposed two-stage recovery strategy can significantly improve the cascading failure resilience of the automotive manufacturing supply chain network.

1. Introduction

The rapid growth of the automotive manufacturing supply chain has been driven by increasing globalization of trade [1,2]. This trend has led to the geographical dispersion of member enterprises across the world and facilitated closer business relationships within the supply chain, resulting in enhanced operational efficiency. However, this interconnectedness also means that the global automotive manufacturing supply chain is vulnerable to various risks and disruptions due to unforeseen events [3–5]. For example, the Tohoku earthquake in Japan in 2011 caused production interruptions for Japanese automotive part manufacturers, which had a ripple effect throughout Europe and North America, impacting the entire global automotive manufacturing supply chain. It is essential to note that the automotive supply chain network comprises numerous enterprises, involving hundreds or even thousands

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of suppliers, and encompasses aspects such as product distribution, inventory management, and sales. Thus, any disruption at any stage can severely affect the smooth operation of the entire supply chain [6–8]. To ensure the stable functioning of the entire supply chain system, it is crucial to conduct extensive research into cascading failures within the automotive manufacturing supply chain network and implement measures to mitigate associated risks.

In recent times, the exploration of cascading failures in supply chain networks has gained significant attention within the research community. These investigations encompass a wide range of network types, such as maritime supply chains, vaccine supply chains, agri-food supply chains, assembly supply chains, plastic waste trading networks, and general supply chains. For instance, Liu et al. [9] developed a framework aimed at identifying and analyzing vulnerable nodes and links in maritime supply chains, with the goal of bolstering the resilience of these networks. Hao et al. [10] employed complex network theory to construct a global lithium supply chain network and simulated the resilience of the system under random and malicious attacks against cascading failures with a nonlinear load-capacity model. Shi et al. [11] studied the resilience of a two-layer interdependent supply chain network composed of an undirected cyber-network and a directed physical-network against cascading failures. In their study, they considered the impact of both functional and structural cascading failures on the network. Li et al. [12] utilized a cascading failure model to assess the breakdown of global plastic waste trade networks. Their findings reveal that 183 countries become overwhelmed as this breakdown propagates, and nearly 6% of countries endure loads exceeding 1.5 times their usual capacity. Hossain et al. [13] proposed an interdependent network model that considered the coupling between port networks and maritime supply chain networks, evaluating the impact of port disruptions on the latter. Golan et al. [14] focused on modeling and assessing the resilience of vaccine supply chain networks, incorporating a time dimension into the failure propagation process by accounting for the patient recovery period. Li et al. [15] introduced the concept of weak ties in cascading failure propagation within agri-food supply chain networks, prioritizing the restoration of supply and demand relationships with neighboring nodes to recover from disruptions. Tang et al. [16] developed a risk propagation cascading failure model for assembly supply chain networks, utilizing production capacity as a resilience indicator and evaluating network resilience under various disruption scenarios. Geng et al. [17] constructed a stochastic-based cascading failure model to quantify the collapse process of the global plastic waste trade network. Gomez et al. [18] developed a cascading failure model for multi-layer supply chain networks, finding that the impact of economic shocks varied based on geographical location and economic sector. Goldbeck et al. [19] proposed a multi-stage stochastic planning model for supply chain networks, considering capacity planning, operational adjustment, and node recovery as decision variables. In our previous work [20], we delved into the cascading failure process within cyber-manufacturing supply chain networks from the perspective of information-services coupling. Our experimental findings demonstrated that optimizing the dependency patterns of information and service nodes effectively enhances the network's resilience against cascading failures.

Though significant progress has been made in the study of cascading failures in supply chains, research on automotive manufacturing supply chain networks remains scarce. In comparison to other supply chain networks, there are notable differences that set the automotive manufacturing supply chain network apart. Firstly, the automotive manufacturing supply chain network involves hundreds or even thousands of member enterprises, with a diverse range of components required for each automotive brand and model. This results in a much larger number of suppliers within the automotive supply chain network. Secondly, the various types of member enterprise nodes within the automotive manufacturing supply chain network differ significantly in size, function, and role. Additionally, automotive manufacturers have the autonomy to select quality parts suppliers and control costs, thus playing a dominant role in the network. These three distinct characteristics make it difficult for existing models to accurately characterize cascading failures in automotive manufacturing supply chain networks.

For the reasons stated above, this study aims to comprehensively model and effectively mitigate cascading failures within automotive manufacturing supply chain networks. To achieve this goal, we initially analyze the roles played by enterprises and develop a practical cascading failure network model tailored specifically to automotive manufacturing supply chains. Subsequently, we propose a two-stage recovery strategy designed to bolster network resilience against cascading failures. Finally, we conduct thorough case studies to investigate the characteristics of cascading failures within the automotive manufacturing supply chain network. The main contributions of this research can be summarized as follows:

(1) Taking into account the varying importance of different enterprise nodes and their loads within real automotive manufacturing supply chain networks, we construct a cascading failure model specifically designed for automotive manufacturing supply chain networks.

(2) Acknowledging the diverse failure states of enterprise nodes, we propose a two-stage recovery strategy to enhance network resilience. In the first stage, underload and attenuated enterprise nodes are prioritized for recovery, followed by the restoration of isolated enterprise nodes in the second stage.

(3) Through comprehensive experiments, we explore the impact of modeling parameters, recovery strategy-related parameters, and different attack strategies on the cascading failure resilience of automotive manufacturing supply chain networks. Furthermore, we validate the effectiveness of the proposed two-stage recovery strategy.

The remainder of this paper is organized as follows. Section 2 introduces the cascading failure model in detail. In Section 3, the two-stage recovery strategy is presented. Section 4 presents the experiment results. Finally, the paper concludes and provides lessons learned.

2. Cascading failure model

2.1. Network model

The automotive manufacturing supply chain network is an intricate and expansive system, resembling a network of interconnected enterprises linked by buyer–supplier connections. Within this network, various node roles exist, including part suppliers,

Table 1
Parameters used in the proposed cascading failure model.

Notation	Description
N	Number of enterprise nodes in the network.
$x_i(t)$	The eigenvector centrality of node i at moment t.
$X(t)$	The eigenvector centrality of all enterprise nodes in the network at time t.
A	The adjacency matrix of the network.
$k_i(t)$	The degree of node i at time t.
$\Gamma_i(0)$	The set of neighboring nodes of node i at time t.
$\delta_{i,j}$	Connection strength of node i and node j at time t.
θ	The load-exponential coefficient.
α	The capacity lower-limit coefficient.
α_i	The lower limit coefficient of node i.
s	Enterprise node types, $s = 1, 2, 3, 4, 5$.
$\overline{k_s(t)}$	The average degree of all s-type nodes at the initial moment.
$\Gamma_i^U(t)$	The set of upstream neighboring nodes of node i at time t
$\Gamma_i^D(t)$	The set of downstream neighboring nodes of node i at time t
β	The attenuation-tolerance coefficient.
$e_i(t)$	The efficiency of node i at time t.
$E(t)$	The relative efficiency of the network at time t
$f_i(t)$	The attenuation degree of node i at time t.

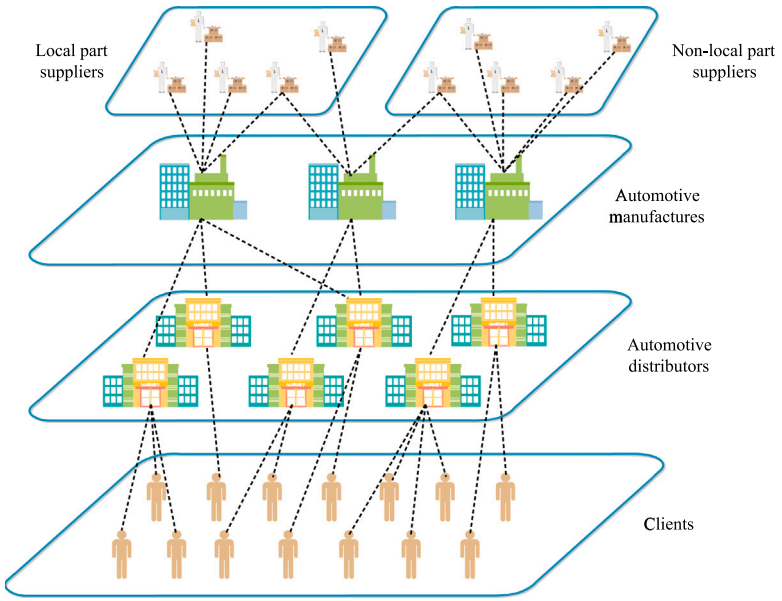


Fig. 1. The layered architecture of the automotive manufacturing supply chain network.

automotive manufacturers, automotive distributors, and clients. Due to the growing global trade, automotive manufacturers now collaborate with numerous part suppliers worldwide. As a result, we introduce the notion of local and non-local part suppliers to account for the geographical distribution of enterprise nodes. Moreover, our automotive manufacturing supply chain network model incorporates the following considerations: (1) there are five distinct node roles based on the proportion: local part suppliers, non-local part suppliers, automotive manufacturers, automotive distributors, and clients; (2) members within the same role do not engage in business relationships, meaning local part suppliers have no connection with non-local part suppliers; (3) parts suppliers exclusively establish business relationships with automotive manufacturers, while automotive manufacturers can engage with clients throughout the hierarchy; (4) the automotive manufacturing supply chain network operates under a client-driven approach, with the automotive manufacturer serving as the central entity and part suppliers and automotive distributors as key members. Therefore, with full consideration of the above four real characteristics, we model the automotive manufacturing supply chain network as a multi-layered network, layered by node roles, as shown in Fig. 1.

Within an actual automotive manufacturing supply chain network, robust enterprises tend to strengthen further, while weaker ones gradually face elimination from the network based on merit. Furthermore, newly established enterprises typically seek to form

business ties with more robust entities. As a result, the network illustrates a pattern where a small set of central enterprise nodes hold the majority of connections, with most other enterprise nodes linked to these central hubs. This configuration highlights the scale-free nature of the real automotive manufacturing supply chain network, characterized by a power-law distribution in degree. We denote the node types as $s = \{1, 2, 3, 4, 5\}$ for local suppliers, non-local suppliers, manufacturers, distributors, and clients, with the degree k_s following a power-law distribution for each type s. Additionally, we depict the network as undirected, and the flow of goods is dictated by the roles of the nodes, moving exclusively from upstream to downstream nodes.

Cascading failures in network systems usually occur when nodes become overloaded. However, in supply chain networks, cascading failures primarily arise from nodes being underloaded due to inadequate demand and supply. Hereafter, we introduce a comprehensive model for cascading failures in the automotive manufacturing supply chain network, encompassing four key aspects: the load-capacity model, the load redistribution process, the cascading failure process, and the resilience metric. The parameters employed in this proposed cascading failure model are consolidated in Table 1.

2.2. Load-capacity model

Within the automotive supply chain network, an enterprise's operational scale is closely linked to its number of business relationships and the profits it generates. Therefore, we define the load of an enterprise node as the operational scale of the participating supply chain members. While prior studies [21–24] typically establish the initial load of a node based on its degree value or betweenness value, the operational scale in a real automotive manufacturing supply chain network is influenced by three main factors: (1) the quantity of upstream and downstream business relationships; (2) the significance of these connections within the entire automotive manufacturing supply chain network; and (3) the strength of the ties with upstream and downstream nodes. The eigenvector centrality considers that a node's significance depends on both the number of its neighboring nodes and the importance of each neighboring node. Consequently, our model gauges an enterprise node's significance in the network based on its eigenvector centrality. A higher eigenvector centrality value indicates a greater importance of the node within the network. Assuming the eigenvector centrality of node i at time t is $x_i(t)$, and $X(t) = [x_1(t), x_2(t), \ldots, x_N(t)]$ represents the eigenvector centrality of all N nodes in the network at time t, and $A = \left(a_{i,j}\right)_{N\times N}$ is the adjacency matrix of the network, then we have

$$x_i(t) = c\sum_{j=1}^{N}(a_{i,j}x_j(t)), \tag{1}$$

where c is a proportionality constant that can be expressed as $X(t) = cAx(t)$ when the network reaches the steady state. The principal unit centroid vector corresponding to the largest eigenvalue of the matrix A is denoted as $X(t)$, and N represents the total number of enterprise nodes in the network. Based on this, the load on node i can be defined as

$$L_i(0) = x_i(0)\sum_{j\in\Gamma_i(0)}\delta_{i,j}(0)^{\theta}, \tag{2}$$

where $x_i(0)$ denotes the eigenvector centrality of node i at the initial moment; $\Gamma_j(0)$ is the set of neighboring node j of node i at the initial moment; $\delta_{i,j}(0)$ is the connection strength between node i and node j at the initial moment; $\delta_{i,j}(t) = k_i(t)\,k_j(t)$, $k_i(t)$ is the degree of node i at time t; θ is the load-exponential coefficient used to control the strength of load on nodes.

An enterprise node's production and supply capacity is constrained by its operational scale in the real world. The load of an enterprise node must never exceed the upper limit of the enterprise's capacity at full capacity. In the automotive manufacturing supply chain, the objective is to provide clients with products. If an enterprise's product demand or raw material supply falls below a certain level, it cannot function normally. Consequently, to operate normally, an enterprise's load must surpass a specific level known as the capacity lower limit. In existing cascading failure models for supply chain networks [15,16], the capacity lower limit is hypothesized to be positively linked to the initial load of nodes. However, due to enterprise heterogeneity, the proportional relationship between the capacity lower limit and initial load differs across all enterprise nodes in automotive manufacturing supply chain networks. Furthermore, adaptability is a crucial factor that influences the capacity lower limit of enterprise nodes. The more adaptable an enterprise node is, the more resilient it becomes to unexpected network events. Degree reflects the number of business relationships that an enterprise node maintains. The higher the degree of an enterprise node, the more business relationships it has. An enterprise node with numerous business relationships is stronger in building connectivity with nodes in other roles, as well as possessing more influence and control over resource allocation and information dissemination. Hence, in our model, we assume that nodes with a higher degree also possess a higher level of adaptability. Li et al. [15] consider node heterogeneity while defining capacity in their work. We employ a similar method to establish the capacity lower limit of node i.

$$C_i = \alpha_i L_i(0), \tag{3}$$

where α_i is the capacity lower-limit coefficient. When node i is an s-type node, its capacity lower-limit coefficient α_i is

$$\alpha_i = \min\{\frac{k_i(0)}{\overline{k_s(0)}}\alpha, \alpha\}, \tag{4}$$

where $\overline{k_s(0)}$ is the average degree of all s-type nodes at the initial moment; $k_i(0)$ is the degree of node i at the initial moment; α is the capacity lower-limit coefficient that satisfies $0 < \alpha < 1$.

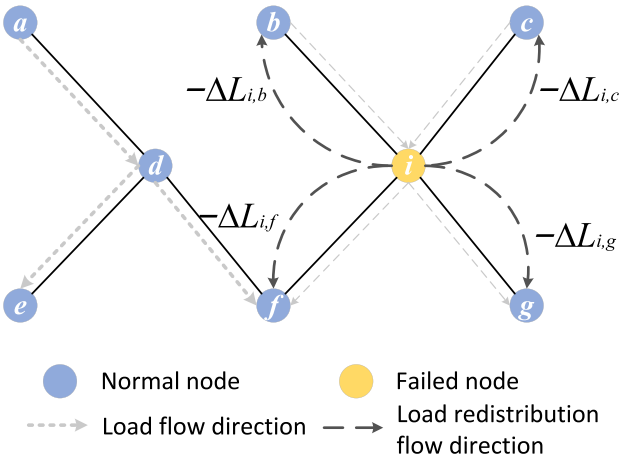


Fig. 2. An example of load redistribution.

2.3. Load redistribution scheme

When an enterprise node in an automotive manufacturing supply chain network fails, it has a ripple effect on its upstream and downstream enterprises. The operational scale of upstream enterprises will decrease due to a drop in product demand, while the operational scale of downstream enterprises will also decline due to a shortage of raw material supply. Consequently, a cascading failure event is triggered. With our prior analysis, we can deduce that the failure of an enterprise can lead to the redistribution of network resources and potentially trigger cascading failure events. The greater the business relationship with the failed enterprise, the more significant the loss for the enterprise. Therefore, our model redistributes the load based on the strength of business relationships between enterprises, known as connection strength. Assuming node i fails at time t, its upstream neighbor nodes b and c, as well as its downstream neighbor nodes f and g, will all suffer losses, as depicted in Fig. 2. As an example, considering the upstream neighbor node b and the downstream neighbor node f, their reduced loads at time t are given by Eqs. (5) and (6), respectively.

$$\Delta L_{i,b}(t) = L_i(t)\frac{\delta_{i,b}(t)}{\sum_{c\in\Gamma_i^U(t)}\delta_{c,i}(t)+\sum_{g\in\Gamma_i^D(t)}\delta_{i,g}(t)}, \tag{5}$$

$$\Delta L_{i,f}(t) = L_i(t)\frac{\delta_{i,f}(t)}{\sum_{c\in\Gamma_i^U(t)}\delta_{c,i}(t)+\sum_{g\in\Gamma_i^D(t)}\delta_{i,g}(t)}, \tag{6}$$

where $\Gamma_i^U(t)$ represents the set of upstream neighboring nodes of node i at time t; $\Gamma_i^D(t)$ represents the set of downstream neighboring nodes of node i at time t.

2.4. Node states

Most existing models for supply chain networks simplify the states of nodes into two categories: normal state and underload state [24,25]. In these models, nodes in the normal state have a load above the lower limit of capacity and can operate without issues. Nodes in the underload state have a load below the lower limit of capacity and cannot function properly. However, in real automotive manufacturing supply chain networks, there can be isolated enterprises resulting from cascading failures. A complete automotive manufacturing supply chain involves various players such as parts suppliers, automotive manufacturers, automotive distributors, and clients, all of which are essential for the manufacturing and delivery process. If any of these players are missing, the automotive manufacturing supply chain network will not operate effectively. Existing graph-based cascading failure models define isolated nodes as nodes that do not belong to the maximum connected component [26,27]. A connected component represents a set of nodes that are interconnected, where any two nodes within the set can be reached by paths. The maximum connected component is the one with the highest number of nodes among all the connected components. From our previous analysis, we deduce that a functioning automotive manufacturing supply chain network requires nodes representing parts suppliers, automotive manufacturers, automotive distributors, and clients simultaneously. Hence, in our model, if a connected component lacks any of these role nodes, all nodes within that connected component should be considered as isolated nodes. Additionally, both local and non-local suppliers are essentially suppliers, so the presence of either type does not affect the automotive manufacturing supply chain.

As depicted in Fig. 3, we can observe various scenarios involving isolated nodes. In Case 1, the failure of a node results in the original connected component transforming into Connected Component 1, while the supply chain continues to operate normally. In Case 2, the failure of a node leads to the original connected component becoming Connected Component 1, causing isolated failures within the nodes of Connected Component 1 due to the absence of component supplier nodes in the network. Case 3

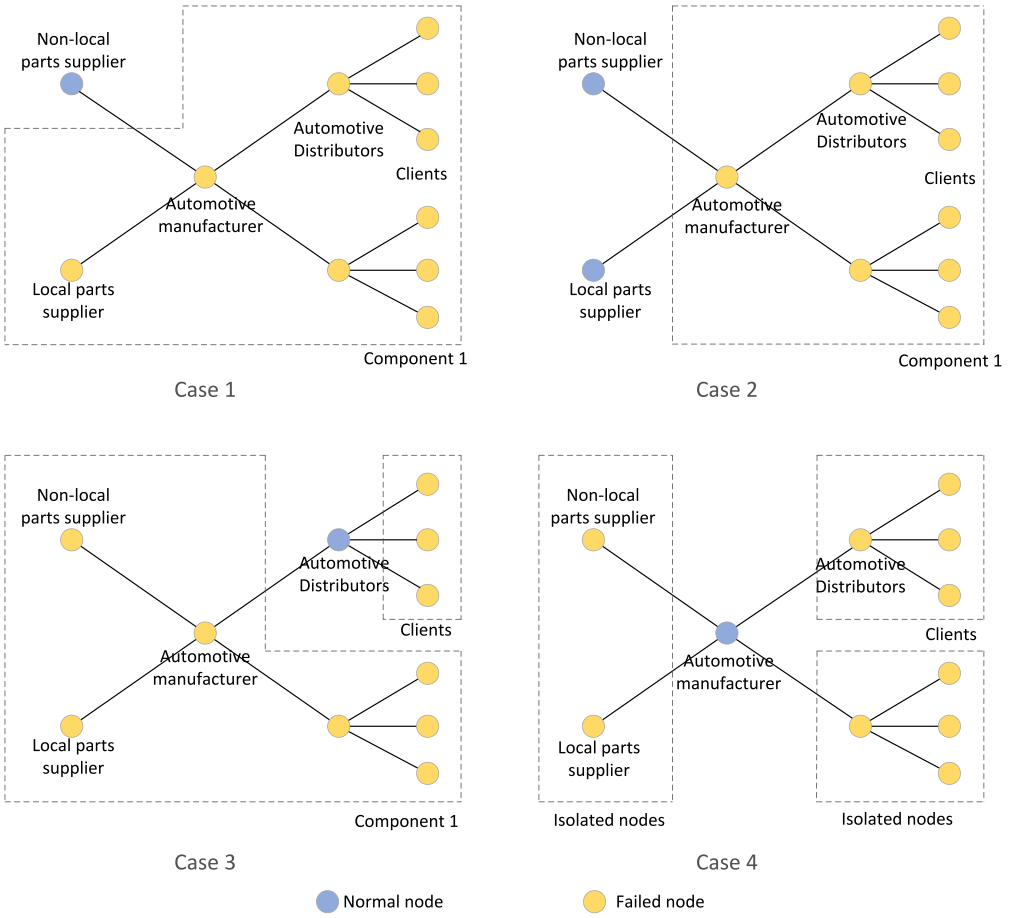


Fig. 3. An example of isolated nodes.

demonstrates that when a node fails, the original connected component becomes Connected Component 1, causing certain nodes to become disconnected and lose their business connections. These disconnected nodes are also categorized as isolated failed nodes. Nonetheless, Connected Component 1 still encompasses nodes with all necessary roles, enabling it to maintain normal operation. Finally, in Case 4, the failure of the sole automotive manufacturer node within the original connected component results in the failure of all remaining nodes in the original connected component due to isolation.

Furthermore, within the supply chain network, a downstream enterprise's decreased demand or an upstream enterprise's insufficient supply of raw materials may lead to a reduction in the operational scale of an enterprise. This state is referred to as the attenuation state within this model. Enterprises in an attenuation state will have certain effects on their upstream and downstream enterprises. Firstly, the decline in demand from downstream enterprises will result in the inability to sell their products promptly, leading to a disruption in capital flows and hindering the production process. With repeated occurrences of this effect, enterprises lacking sufficient funds to continue production will eventually go bankrupt. Secondly, the insufficient supply of raw materials from upstream enterprises will hinder production activities, forcing enterprises to rely on existing stocks of raw materials and products to sustain their operations. Once these inventories are depleted, the enterprise will face the risk of production shutdown. Therefore, in our model, a node is considered to be in an attenuation state if its load falls below the lower limit of attenuation but remains above the capacity lower limit.

Drawing upon the preceding analysis, our definition of node states diverges from prior studies [24–27]. Our model encompasses four distinct states for enterprise nodes: normal state, attenuation state, underload state, and isolated state. To quantify the degree of attenuation experienced by an enterprise node i at time t, we introduce the concept of an attenuation function denoted as $f_i(t)$:

$$f_i(t) = \begin{cases} 0, & L_i(t) \geq (1+\beta)C_i \\ 1 - \frac{L_i(t)-(1+\beta)C_i}{(1+\beta)C_i - C_i}, & C_i \leq L_i(t) \leq (1+\beta)C_i \\ 1, & L_i(t) \leq C_i, \end{cases} \tag{7}$$

where β is the attenuation-tolerance coefficient. When $L_i(t) \geq (1+\beta)C_i$, $f_i(t) = 0$, the enterprise node i works normally; when $C_i \leq L_i(t) \leq (1+\beta)C_i$, the enterprise node i is in an attenuation state, indicating that the enterprise is still operating, but the scale of

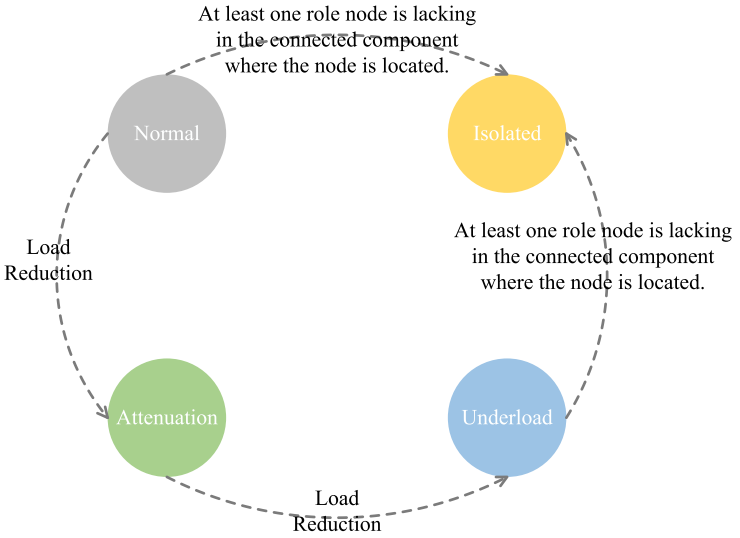


Fig. 4. State transitions of enterprise nodes in the cascading failure process of the automotive manufacturing supply chain network.

its operation has shrunk; when $L_i(t) \leq C_i$, $f_i(t) = 1$, the enterprise node i stops operating due to reasons such as inventory depletion and is in an underload state. Fig. 4 shows the state transitions of enterprise nodes in the automotive manufacturing supply chain network during cascading failures.

2.5. Cascading failure process

In order to illustrate the cascading failure process within our proposed model, Fig. 5 offers a visual depiction of such a process in an automotive manufacturing supply chain network. Within this network, node 1 represents a local parts supplier, node 2 a non-local parts supplier, nodes 3 and 4 are automotive manufacturers, nodes 5 and 6 are automotive distributors, and nodes 7, 8, 9, and 10 denote clients. In Fig. 5(a), the initial state of the network is depicted, demonstrating normal business transactions conducted by each enterprise node and the efficient operation of the network. Subsequently, Fig. 5(b) portrays the scenario where node 4 ceases to function due to an unexpected event, such as an earthquake or strike. This failure impacts nodes 1, 2, 5, and 6, which have business relationships with node 4. Assuming that the load redistribution resulting from the failure of node 4 causes nodes 1, 2, 5, and 6 to fall below their minimum capacity limits, nodes 1, 2, 4, and 6 are classified as failing due to underload. Following the failure of these nodes, their loads are redistributed, further influencing the load states of neighboring nodes 3, 7, 8, 9, and 10. In consideration of the diversity among enterprise nodes, it is assumed that after load redistribution, nodes 7, 9, and 10 also fail due to underload, while nodes 3 and 8 remain operational. This results in the network state depicted in Fig. 5(c). As the connected components containing nodes 3 and 8 lack other essential roles, these two nodes are deemed to fail due to isolation. At this point, all nodes in the network have failed, ultimately leading to the final state following the cascading failure, as illustrated in Fig. 5(d).

To offer a more lucid depiction of the cascading failure propagation process within the automotive manufacturing supply chain network, we have devised a time-varying cascading failure function that builds upon the aforementioned process. In our subsequent analysis, we denote $U(\xi)$ as the ratio of underloaded nodes following load redistribution, while $I(\xi)$ signifies the proportion of isolated nodes within the network. Assuming the removal of a fraction P of nodes, the remaining operational nodes in the network amount to $\xi_1 = N(1-P)$. This node removal triggers a redistribution of loads, which subsequently leads to the underload-induced failure of certain nodes. Consequently, the count of functional nodes in the network becomes $\xi_2' = N - U(\xi_1)$. Subsequent isolated failures within the supply chain prompt the count of operational nodes to shift to $\xi_2 = N - I(\xi_2')$. The failure of isolated nodes initiates another round of load redistribution in the network. As a result of removing the underload-failed nodes, the count of normally functioning nodes becomes $\xi_3' = N - U(\xi_2)$. These underload-induced failures then lead to isolated failures of specific nodes. Upon removing the isolated failed nodes, the tally of normal nodes in the network is $\xi_3 = N - I(\xi_3')$.

According to the above description, we can easily obtain the recurrence relation of cascading failure, and the dynamic function of normally functioning nodes at each stage can be expressed as

$$\begin{aligned} &t = 1 : \xi_1 = N(1-P) \\ &t = 2 : \xi_2{}' = N - U(\xi_1), \xi_2 = N - I(\xi_2{}') \\ &t = 3 : \xi_3{}' = N - U(\xi_2), \xi_3 = N - I(\xi_3{}') \\ &\vdots \\ &t = n : \xi_n{}' = N - U(\xi_{n-1}), \xi_n = N - I(\xi_n{}') \end{aligned} \quad , \tag{8}$$

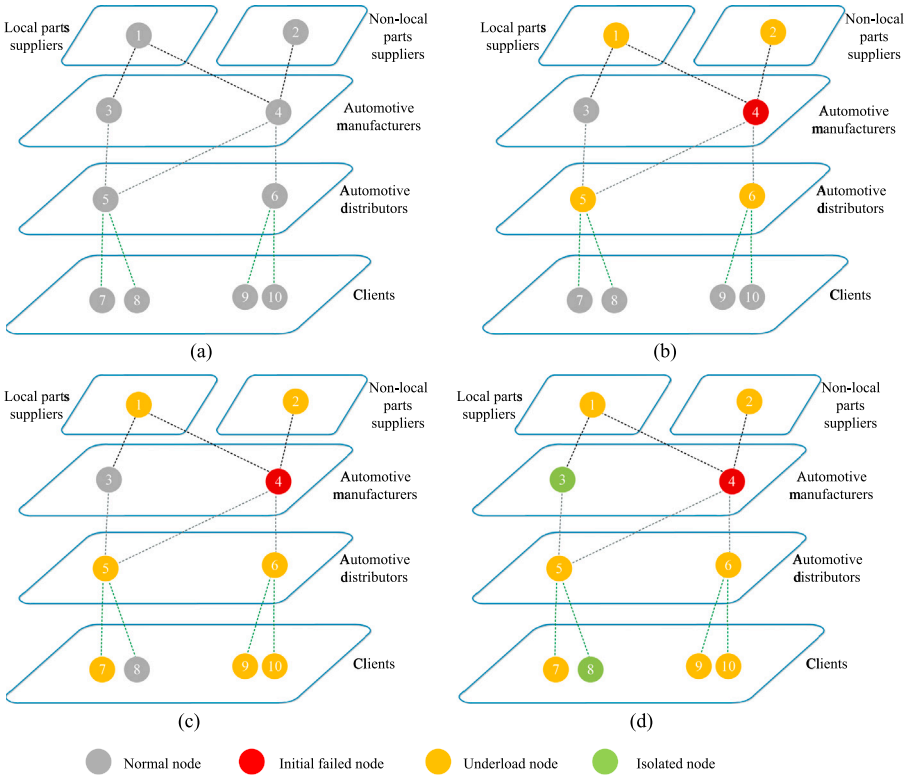


Fig. 5. An example of cascading failure process in automotive manufacturing supply chain network.

when $n \to \infty$, the function reaches a steady state, i.e. $\xi_n = \xi_{n+1}$. This is because at this point, there are no more failed nodes in the network. Therefore, we can get

$$n \to \infty : \xi_n{}' = N - U(\xi_n), \xi_n = N - I(\xi_n{}'). \tag{9}$$

Eventually, the number of nodes that can operate normally in the network is $\xi_n = N - I(\xi_n{}')$.

2.6. Resilience metric

The resistance to damage exhibited by an automotive manufacturing supply chain network is commonly referred to as its resilience. Various metrics have been employed to assess network resilience, including the relative size of the maximum connected component [28] and the relative size of normal nodes [29,30]. However, upon conducting node state analysis in Section 2.4, we discovered that the nodes within the maximum connected component of the automotive manufacturing supply chain network do not necessarily represent the normal nodes in the network. Consequently, the relative size of the maximum connected component cannot serve as a suitable resilience metric for the automotive manufacturing supply chain network. Moreover, the relative size of normal nodes fails to capture the impact caused by enterprises in an attenuated state on the network. Thus, the relative size of normal nodes is also inadequate as a resilience metric for the automotive manufacturing supply chain network. In this study, we propose the utilization of relative network efficiency as a measure of cascading failure resilience in manufacturing supply chain networks. This metric effectively reflects the influence of efficiency changes in each node on the overall network efficiency. In our model, the efficiency of node i at time t is denoted as $e_i(t)$, ranging from 0 to 1. We assume that all nodes in the network initially function properly (i.e., $e_i(0) = 1$). If the load on node i decreases at time t and satisfies the condition $L_i(t) \geq (1+\beta)C_i$, then node i remains in a normal state. However, when $L_i(t) < (1+\beta)C_i$, node i enters the attenuation state at that moment, and its efficiency at time t is

$$e_i(t) = \frac{L_i(t) - C_i}{(1+\beta)C_i - C_i}. \tag{10}$$

According to (10), $e_i(t) = 0$ indicates that the node is in an isolated or underload state, $e_i(t) = 1$ indicates that the node is in a normal state, and $e_i(t) \in (0, 1)$ indicates that the node is in an attenuation state. On this basis, the relative network efficiency $E(t)$ is defined as the average value of all enterprise nodes in the automotive manufacturing supply chain network at time t, that is

$$E(t) = \sum_{i=1}^{N} \frac{e_i(t)}{N}. \tag{11}$$

Table 2
Parameters in the two-stage recovery strategy.

Notation	Description
Δt	Recovery time.
p	The underload/attenuation state-transition coefficient.
$p_i^U(t)$	The recovery probability of underload node i at time t.
$p_i^A(t)$	The recovery probability of attenuation node j at time t.
$F_i^U(t)$	The underload state transition function.
$F_i^A(t)$	The attenuation state transition function
r	The isolated state-transition coefficient.
$r_{j,i}(t)$	The priority of node j in establishing a business relationship with node i.
$R_{j,i}(t)$	The isolated state transition function of node i.

When $t \to \infty$, the network reaches a steady state, and the relative network efficiency at this state is denoted as E_{end}. We choose E_{end} as the final metric for evaluating the resilience of automotive manufacturing supply chain networks against cascading failures.

3. Resilience recovery strategy for automotive manufacturing supply chain network

3.1. Two-stage recovery strategy

Real-world automotive manufacturing supply chain networks exhibit four states for nodes following cascading failures: isolated, underload, attenuated, and normal. Failed and attenuated nodes can be restored to their normal state through the implementation of appropriate recovery measures within a designated timeframe. In this study, we put forth a two-stage recovery strategy that entails distinct recovery measures for attenuated nodes and various types of failed nodes in each stage.

In the event of a significant incident, such as parts shortage, causing an enterprise to experience attenuation or underload, specific approaches are implemented to reinstate the original normal state. The likelihood of recovering attenuated nodes is expected to be greater than that of underload nodes due to less severe impairment. For instance, when Shanghai experienced a widespread outbreak in March 2022, automobile manufacturers, including NIO, Tesla, and SAIC, halted production due to part shortages, leading to underload. As the situation gradually improved, SAIC was the first to resume work and production in April, followed by Tesla and NIO. Given the varying degrees of importance of diverse enterprises within the supply chain network and their ability to adapt to unexpected events, their capacity to recover and restore to a normal state can differ at various recovery times. Those underload enterprises with high significance and adaptability tend to demonstrate better resilience and can more effectively restore to a normal state.

In order to enhance the practicality of the recovery strategy, we introduce a modification whereby the sequence of recovery processes for isolated failed nodes is adjusted to follow the recovery processes for underload nodes and attenuation nodes. When the most upstream node within a connected component does not serve as a part supplier node and its original upstream node has been successfully recovered, the connected component containing the original isolated failed node will establish a connection with the recovered node from the initial linkage. In cases where the original upstream node remains unrecovered, the node will establish a business relationship with an upstream node that possesses a comparable operational scale and a high level of compatibility. On the other hand, if the most upstream node within a connected component functions as a supplier node and its original downstream node has been restored, the connected component housing the original isolated failed node will be reconnected with the initially recovered node. Consequently, a two-stage recovery strategy is proposed in this study. The first stage involves restoring underload enterprise nodes to their normal operational state, while the second phase focuses on establishing new business relationships for the isolated failed enterprise nodes. Detailed parameters describing the proposed two-stage recovery strategy are provided in Table 2.

In the event of cascading failures, the failed nodes and attenuation nodes have a probability of recovery at each interval Δt. Δt is the recovery time, which is an adjustable coefficient. The two-stage recovery strategy is as follows:

First stage: Suppose that node i in the network is in an underload state at time t. The load of this node is $L_i(t)$, and the recovery coefficient $p_i^U(t)$ of this underload node is

$$p_i^U(t) = \frac{C_i}{L_i(0)} x_i(t). \tag{12}$$

The underload state transition function $F_i^U(t)$ is used to determine whether node i reverts to its original state, that is

$$F_i^U(t) = p_i^U(t) - p, \tag{13}$$

where p is the underload/attenuation state-transition coefficient, satisfying $0 < p < 1$. If $p_i^U(t) > p$, i.e., $F_i^U(t) > 0$, node i is restored to normal state; if $p_i^U(t) \le p$, i.e., $F_i^U(t) \le 0$, node i is still in the failure state.

Suppose that node j in the network is in an attenuation state at time t. The load of this node is $L_j(t)$, and the recovery coefficient of this attenuation node $p_j^A(t)$ is

$$p_j^A(t) = \frac{L_j(t)}{L_j(0)} x_j(t). \tag{14}$$

The attenuation state transition function $F_j^A(t)$ is used to determine whether node j reverts to its original normal state, that is

$$F_j^A(t) = p_j^A(t) - p. \tag{15}$$

If $p_j^A(t) > p$, i.e., $F_j^A(t) > 0$, node j is restored to normal state; if $p_j^A(t) \le p$, i.e., $F_j^A(t) \le 0$, node j is still in the attenuation state.

Second phase: If the original upstream and downstream nodes of the isolated node are restored in the first phase, it will establish connections with the originally connected nodes. If the original upstream and downstream nodes of the isolated node are not restored, the node will establish a business relationship with another node with a similar scale of operation and high adaptability in the upstream and downstream. This prioritization aims to restore the network as closely as possible to its initial state. Suppose node i fails at time t. It will select node j whose load $L_i(t)$ is the closest and has a high degree of adaptation to establish the business relationship. We use $r_{j,i}(t)$ to measure the priority of node j in establishing a business relationship with node i:

$$r_{j,i}(t) = \frac{k_j(t)}{1 + |L_j(t) - L_i(t)|}. \tag{16}$$

The isolated state transition function $R_{j,i}(t)$ is used to determine whether node i establishes a new business relationship with node j at time t:

$$R_{j,i}(t) = r_{j,i}(t) - r, \tag{17}$$

where r is the isolated state-transition coefficient, satisfying $0 < r < 1$. If $r_{j,i}(t) > r$, i.e., $r_{j,i}(t) > 0$, node i will establish business relationship with node j. If $r_{j,i}(t) \le r$, i.e., $r_{j,i}(t) \le 0$, node i will not establish business relationship with node j.

The automotive manufacturing supply chain network is repaired through the above two phases after cascading failures. If there are still failed nodes in the network after one round of repair, a new round of cascading failures maybe triggered again. The network will only reach a steady state when no new failed nodes appear in the network. An network cascading failure propagation process after adopting the two-stage recovery strategy is shown in Fig. 6.

3.2. Example of recovery process under two-stage recovery strategy

Consider the scenario depicted in Fig. 7, representing the recovery process of an automotive manufacturing supply chain network. We assume a recovery time $\Delta t = 1$. At time t, the network state after cascading failures is as follows: Nodes 2, 3, and 5 are experiencing underload, while nodes 4, 6, 7, 8, 9, 10, and 11 are isolated. Node 1 initially failed. The recovery process begins at time $t + 1$ with the first stage aimed at restoring underload nodes, initially failed nodes, and attenuation nodes in the network. Assuming that transition functions $F_1^U(t+1)$ and $F_3^U(t+1)$ for nodes 1 and 3 are greater than 0, nodes 1 and 3 return to their normal operating states, along with the restoration of original connections between them. However, the transition function $F_2^U(t+1)$ of node 2 is less than 0, indicating that it remains in a failed state. Consequently, the network topology at the end of the first stage appears as depicted in Fig. 7(b). The second stage of the recovery strategy commences, wherein the isolated nodes 4, 6, 7, and 8 form a connected component, and nodes 9, 10, and 11 remain isolated. The most upstream node, node 4, prioritizes establishing business relationships with other nodes. As node 3, originally in a business relationship with node 4, has returned to its normal state, node 4 takes precedence in establishing a business relationship with node 3. In the absence of restored upstream nodes for nodes 9, 10, and 11, they choose node 4, which is closest to their load and highly adaptable, as their business partner. Suppose $R_{9,4}(t+1)$ is greater than 0 and $R_{10,4}(t+1)$ and $R_{11,4}(t+1)$ are less than 0. Consequently, node 9 establishes a new business relationship with node 4, whereas nodes 10 and 11 remain isolated. At the end of the second stage, the recovery process concludes, and no additional failed nodes emerge. The entire network reaches its final steady state, as depicted in Fig. 7(c).

4. Experimental results

4.1. Practical case

The automotive manufacturing supply chain operates through a multi-tiered system of parts suppliers, comprising first-tier, second-tier, and third-tier suppliers. First-tier suppliers play a crucial role in designing and manufacturing automotive component modules such as seats and instrument panels. Second-tier suppliers specialize in providing core technologies, like chips, while third-tier suppliers primarily focus on supplying raw materials essential for automotive manufacturing, such as steel. Our case model primarily focuses on first-tier suppliers involved in automotive manufacturing, not limited to those based in China. Additional suppliers from overseas contribute to the supply chain by providing high-quality component modules, including body parts, engines, and electrical equipment. Consequently, the automotive manufacturing supply chain network revolves around auto manufacturers and encompasses various entities, including local and non-local first-tier suppliers, manufacturers, distributors, and clients.

To enhance the alignment of experimental results with real-world scenarios, we utilized data from a comprehensive study [31] and drew on research findings from the top twelve automotive brands in the Chinese market spanning the last five years. From this pool of brands, we meticulously selected the most emblematic enterprises to construct an automotive supply chain network. The model was initialized with 12 automotive brands and 32 automotive manufacturers within the network, each linked to a single distributor for every automotive brand across China's 34 provincial-level administrative regions, resulting in a total of 408 automotive distributors. Assuming an average of 10 domestic suppliers for each type of component module, there are 40 local suppliers for the four types of component modules mentioned earlier in China. Considering the relatively limited engagement

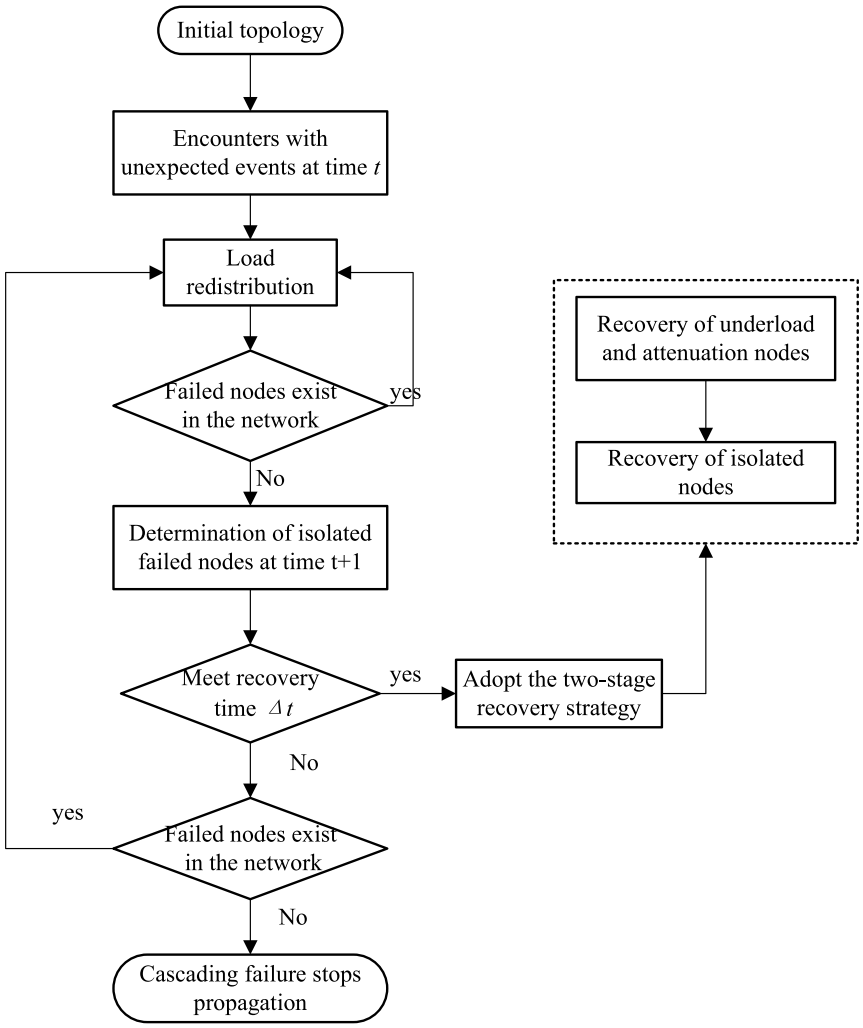


Fig. 6. Cascading failure propagation process after adopting two-stage recovery strategy.

of foreign component module suppliers with domestic manufacturers, we estimated an average of 6 suppliers for each type of component module, totaling 24 non-local suppliers. Within the identified enterprises, automotive manufacturers and parts suppliers were classified into three distinct categories: A, B, and C. Class A denotes enterprises with substantial scale and robust international business capabilities, class B represents enterprises with significant scale and strong domestic business capabilities, while class C encompasses smaller-scale enterprises with average business capabilities. Clients were classified into three categories based on their purchasing proclivity: those with a strong willingness to buy, more interested clients, and ordinary clients, with enterprises and clients in each category conforming to a power-law distribution. For the purpose of simulation analysis, we carefully selected the most representative 200 enterprises and client agents, culminating in a supply chain composition of 20 local suppliers, 10 non-local suppliers, 15 manufacturers, 55 distributors, and 100 client agents.

4.2. Experimental setup

To conduct our simulation experiments, we employed the MATLAB software. In order to expedite the simulation process, we leveraged the GUROBI optimization tool integrated within the platform. The experiments were conducted with the following configurations: (1) every enterprise node in the initial network was assumed to be operational, and the network topology featured a complete interconnection; (2) for the single-node attack scenario, each node was sequentially targeted, and measurements were averaged accordingly; (3) for the multi-node attack scenario, we selected the top 5% of enterprise nodes and subjected them to various attack strategies aimed at triggering cascading failures. All experimental outcomes were derived from an average of 50 interdependent simulations.

Based on the practical case presented in section 4.1, we conducted a series of simulation experiments to explore various aspects: (1) We examined the influence of network modeling coefficients, which encompassed the load-exponential coefficient θ, capacity

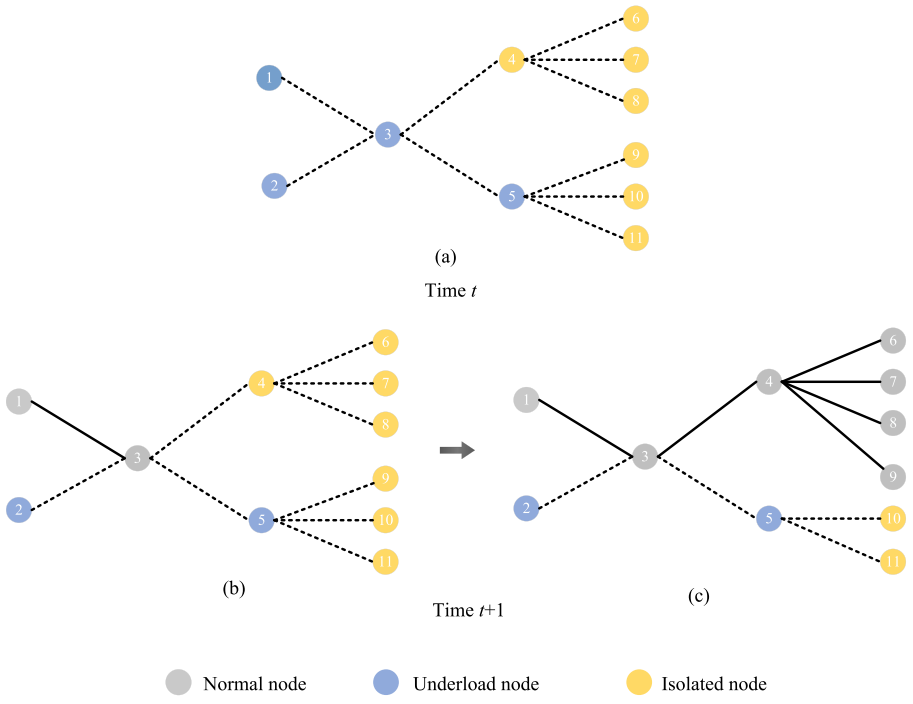


Fig. 7. An example of recovery process of automotive manufacturing supply chain network.

lower-limit coefficient α, and attenuation coefficient β, on the resilience of cascading failures within the network. (2) We analyzed the network's resilience against different attack strategies, including those focusing on high eigenvector centrality, low eigenvector centrality, high-degree nodes, and low-degree nodes. (3) We investigated the impact of recovery strategy-related coefficients, specifically the recovery time Δt, underload/attenuation state-transition coefficient p, and isolated state-transition coefficient r, on the network's ability to withstand cascading failures. (4) We validated the effectiveness of a proposed two-stage recovery strategy. Through these simulation experiments, our goal was to acquire a deeper understanding of the network's dynamics and behaviors. We sought to investigate its ability to withstand cascading failures and assess the effectiveness of different strategies and coefficients utilized.

4.3. Impact of modeling parameters

4.3.1. Impact of load-exponential coefficient θ

Fig. 8 compares the relative network efficiency $E(t)$ of the automotive manufacturing supply chain network with different load-exponential coefficients θ. It can be observed that the relative network efficiency $E(t)$ decreases significantly with the increase of θ. When $\theta = 1$, $E(t)$ is 86%. When θ rises to 4, $E(t)$ decreases to 62%. In the proposed cascading failure model, the load-exponential coefficient θ determines the load size of enterprise nodes, and a higher θ means that the load of enterprise nodes increases exponentially. This leads to a more uneven load distribution of enterprise nodes, aggravating the risk of cascading failures. Therefore, in the automotive manufacturing supply chain network, a smaller θ helps to make the load distribution of enterprise nodes more uniform, thus improving the cascading failure resilience.

Fig. 9 displays the composition of enterprise nodes after the network reaches a steady state at different θ. Through comparative analysis, we can easily observe that the proportion of attenuated enterprise nodes and failed enterprise nodes tends to increase as the load-exponential coefficient θ increases. Furthermore, we can also observe that the proportion of underload failed nodes among the failed nodes is much higher than the proportion of isolated failure enterprise nodes. Therefore, we believe that underload failed nodes are the primary contributing factor to the decrease in cascading failure resilience of the automotive manufacturing supply chain network. This finding highlights the significant impact of load redistribution among failed nodes on the network's cascading failure resilience.

4.3.2. Impact of capacity lower-limit coefficient α

Fig. 10 presents the relative network efficiency $E(t)$ of the automotive manufacturing supply chain network over time with different capacity lower-limit coefficients α. It is evident that as the capacity lower-limit coefficient α increases, the relative network efficiency $E(t)$ exhibits a decreasing trend, indicating a negative correlation between $E(t)$ and α. In the automotive manufacturing supply chain network, node failures occur when the current load of enterprise nodes falls below the capacity lower-limit coefficient.

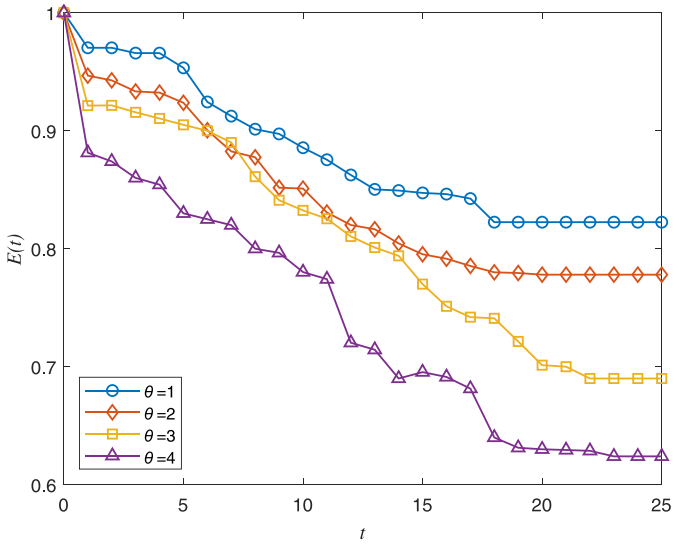


Fig. 8. Relative network efficiency $E(t)$ varying θ ($\sigma = 0.6$, $\beta = 1$).

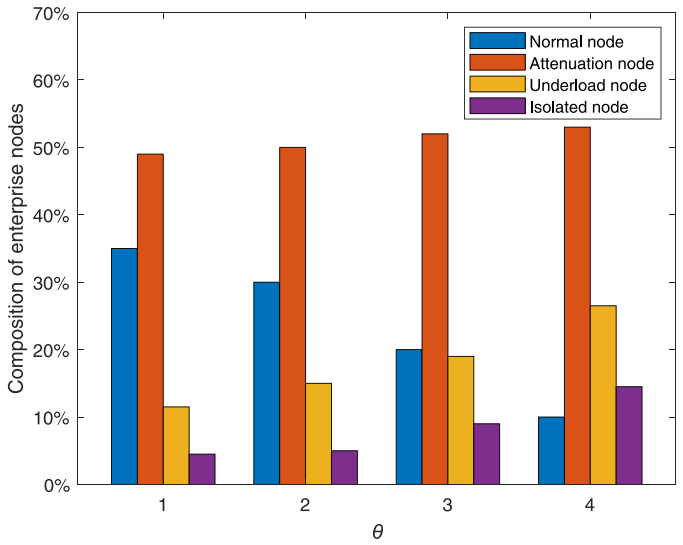


Fig. 9. Composition of enterprise nodes varying θ ($\sigma = 0.6$, $\beta = 1$).

This means that the current product demand or raw material supply of the enterprise is insufficient to generate profit and sustain normal production activities. In some cases, it may even lead to business closure due to an inability to generate profit. A higher capacity lower-limit coefficient signifies lower risk resilience for enterprises in facing cascading failures, thereby increasing the likelihood of failure and causing cascading failures within the automotive manufacturing supply chain network.

Fig. 11 displays the composition of enterprise nodes after the network reaches a steady state at different α. We can observe that the proportion of failed enterprise nodes increases significantly with the increase of α. In our proposed model, the capacity lower-limit parameter determines the profit loss that an enterprise node can withstand. When the incurred loss falls below its capacity lower-limit, the enterprise node fails due to bankruptcy and underload. Therefore, in order to help the automotive manufacturing supply chain network resist cascading failures, enterprise nodes should enhance their capacity to withstand losses.

4.3.3. Impact of attenuation coefficient β

Fig. 12 shows the relative network efficiency $E(t)$ of the automotive manufacturing supply chain network with different attenuation-tolerance coefficient β over time. We can easily observe a significant reduction in the relative network efficiency $E(t)$ as the attenuation-tolerance coefficient β increases. The negative correlation between $E(t)$ and β can be reasonably explained by (7): as the attenuation-tolerance coefficient increases, the network's tolerance for profit loss decreases. This implies that when an enterprise

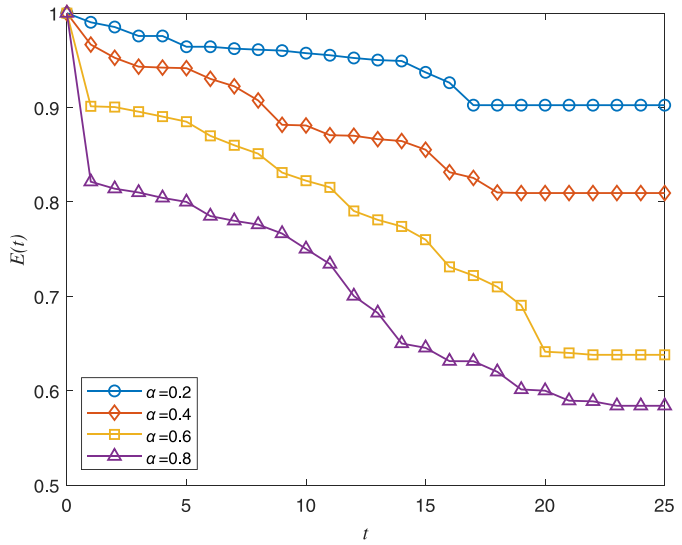


Fig. 10. Relative network efficiency $E(t)$ varying α ($\theta = 2$, $\beta = 0.2$).

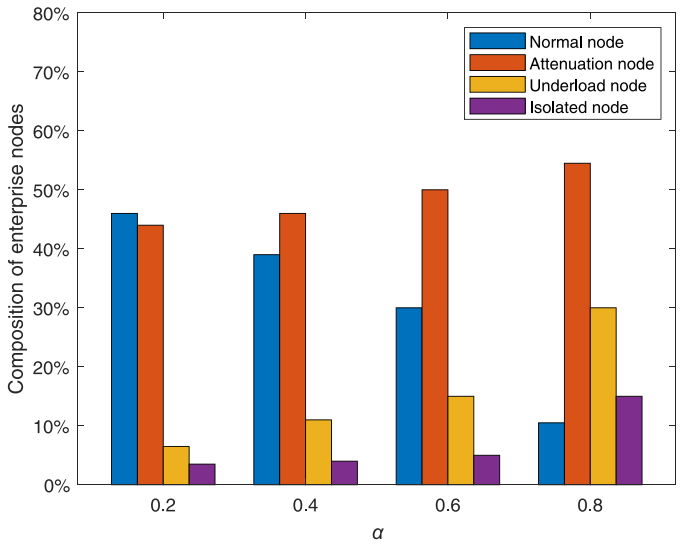


Fig. 11. Composition of enterprise nodes varying α ($\theta = 2$, $\beta = 0.2$).

experiences profit loss at time t, it is more likely to enter the attenuation state, resulting in a decrease in enterprise efficiency and consequently affecting the efficiency of the entire automotive manufacturing supply chain network.

Fig. 13 illustrates the composition of enterprise nodes as β changes. It can be observed that the proportions of attenuation enterprise nodes and failed enterprise nodes exhibit a negative correlation with the load attenuation-tolerance coefficient β. In our proposed model, β determines the load threshold value for nodes to enter the attenuation state. As the value of β increases, the probability of nodes entering the attenuation state after load reduction also increases. Therefore, to enhance the resilience of the automotive manufacturing supply chain network, enterprise nodes should prioritize maintaining an adequate internal inventory level to bolster their capacity for tackling external risks.

4.4. Impact of attack strategies

In order to evaluate the impact of different attack strategies on the cascading failure resilience of the automotive manufacturing supply chain network, we utilize the following four attack strategies that target the top 5% of nodes in the network: (1) High Eigenvector Centrality Attack (HECA), which targets enterprise nodes in descending order of eigenvector centrality; (2) Low

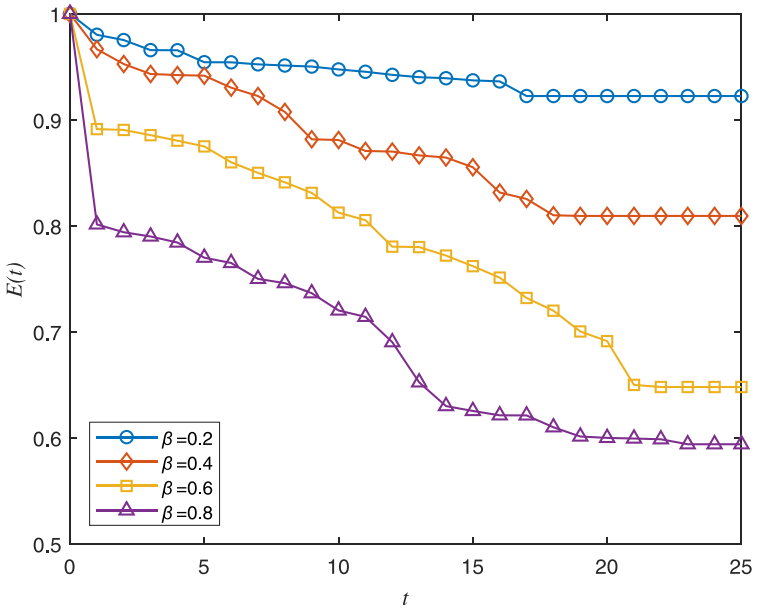


Fig. 12. Relative network efficiency $E_{(t)}$ varying β ($\alpha = 0.6$, $\theta = 2$).

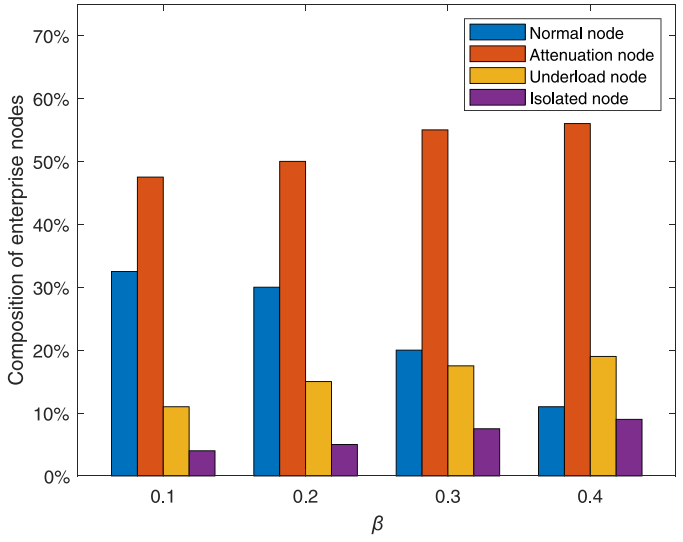


Fig. 13. Composition of enterprise nodes varying β ($\alpha = 0.6$, $\theta = 2$).

Eigenvector Centrality Attack (LECA), which targets enterprise nodes in ascending order of eigenvector centrality; (3) High Degree Attack (HDA), which targets enterprise nodes in descending order of degree; and (4) Low Degree Attack (LDA), which targets enterprise nodes in ascending order of degree.

Fig. 14 displays the relative network efficiency $E(t)$ under different attack strategies. It can be observed that among the four attack strategies, the HECA strategy has the greatest impact on the cascading failure resilience of the automotive manufacturing supply chain network, followed by the HDA strategy, and finally the LECA and LDA strategies.

Fig. 15 illustrates the composition of enterprise nodes under different attack strategies. It is evident that the proportion of nodes experiencing underload failures is significantly higher when utilizing high eigenvector centrality and high-degree attack strategies, compared to the other two attack strategies. This can be attributed to the fact that enterprise nodes with high importance and adaptability tend to have higher loads, leading to a more substantial impact on a greater number of nodes through load redistribution. Based on the findings from the experiments conducted, it is recommended to prioritize high-importance and high-adaptability enterprises over low-importance and low-adaptability enterprises.

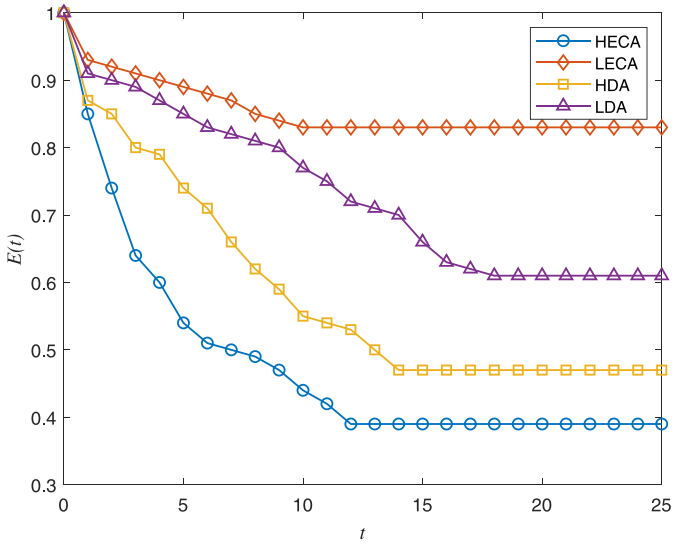


Fig. 14. Relative network efficiency $E_{(t)}$ under different attack strategies ($\alpha = 0.6$, $\sigma = 0.6$, $\beta = 1$).

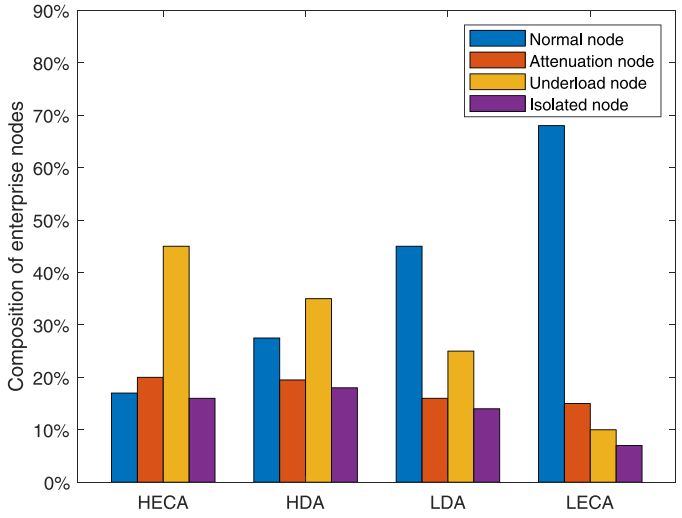


Fig. 15. Composition of enterprise nodes under different attack strategies ($\alpha = 0.6$, $\sigma = 0.6$, $\beta = 1$).

4.5. Impact of recovery strategy-related coefficients

This section explores the impact of recovery strategy-related parameters on the cascading failure resilience of automotive manufacturing supply chain networks. The recovery strategy-related parameters consist of three types: the recovery time Δt; the underload/attenuation state-transition coefficient p; and the isolated state-transition coefficient r.

4.5.1. Impact of recovery time Δt

Fig. 16 compares the steady-state values E_{end} of the relative network efficiency for the automotive manufacturing supply chain network under different recovery times Δt. It is evident that as the recovery time Δt decreases, the relative network efficiency (E_{end}) experiences a significant increase. For example, when $r = 0.4$ and Δt=1, the value of E_{end} is 0.8, whereas when $r = 0.4$ and Δt=5, the value of E_{end} drops to 0.69.

Fig. 17 illustrates the composition of enterprise nodes after the network reaches a steady state at different Δt values. It can be observed that as the recovery time Δt increases, the proportion of normal enterprise nodes decreases while the proportion of failed enterprise nodes increases. This trend occurs because longer recovery times Δt result in more attenuated and failed nodes due to cascading failures. Additionally, during the propagation of cascading failure, attenuated and failed nodes have fewer opportunities

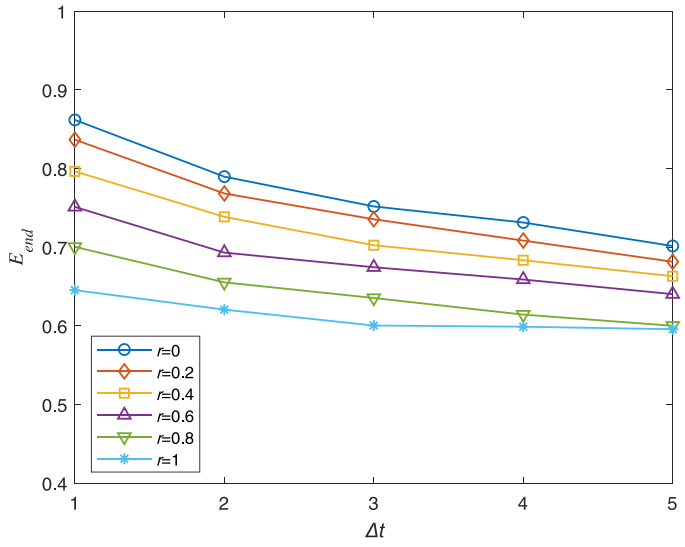


Fig. 16. Relative network efficiency steady-state value E_{end} varying Δt ($\alpha = 0.8$, $\theta = 2$, $\beta = 0.2$, $p = 0.5$).

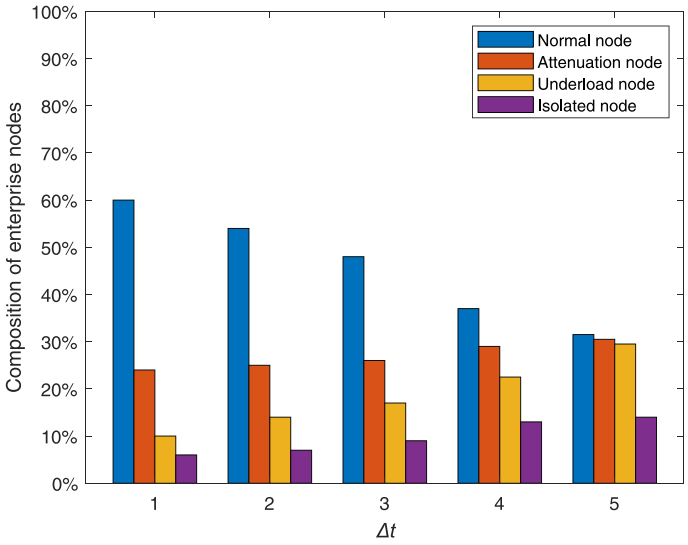


Fig. 17. Composition of enterprise nodes varying Δt ($\alpha = 0.8$, $\theta = 2$, $\beta = 0.2$, $p = 0.5$, $r = 0.5$).

for recovery. Both of these factors contribute to lower cascading failure resilience in the network when employing a longer recovery time strategy. Therefore, in the automotive manufacturing supply chain network, enterprises should strive to enhance recovery efficiency and shorten the recovery time to improve the cascading failure resilience of the network.

4.5.2. *Impact of underload/attenuation state-transition coefficient p*

Fig. 18 illustrates the steady-state value E_{end} of the relative network efficiency in the automotive manufacturing supply chain network with different underload/attenuation state-transition coefficients p. It is evident that there is a negative correlation between the steady-state value of relative network efficiency E_{end} and p. By referring to Eqs. (12), (13), (14), and (15), we can understand the reason behind this negative correlation. A larger value of p implies that the underload state transition function $F_i^U(t)$ or the attenuation state transition function $F_i^A(t)$ is less likely to be greater than 0 at time t. As a result, the failed node i in the underload/attenuation state is less likely to return to its normal operating state, resulting in a smaller E_{end}.

Fig. 19 presents the composition of enterprise nodes after the network reaches a steady state at different values of p. Through comparative analysis, we observe that the proportion of attenuation and underload enterprise nodes increases with the increasing underload/attenuation state-transition coefficient p. Particularly, the number of underload failed nodes shows a more pronounced increase. In the two-stage recovery strategy, the adjustable coefficient p is utilized as the criterion for determining when an

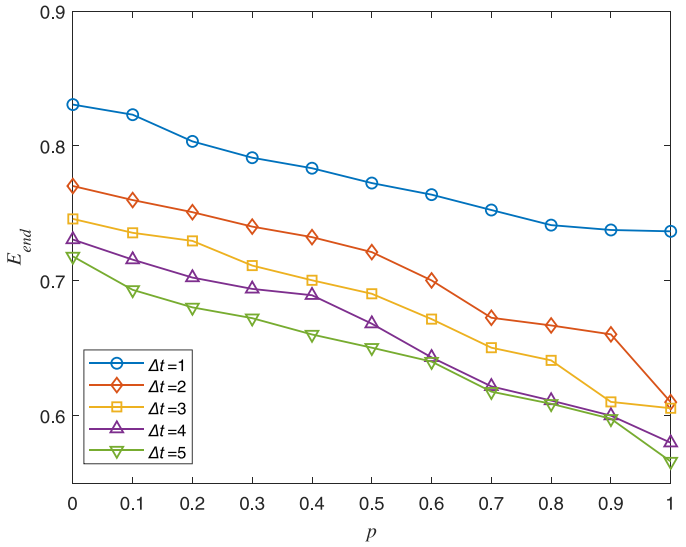


Fig. 18. Relative network efficiency steady-state value E_{end} varying p ($\alpha = 0.8$, $\theta = 2$, $\beta = 0.2$, $r = 0.5$).

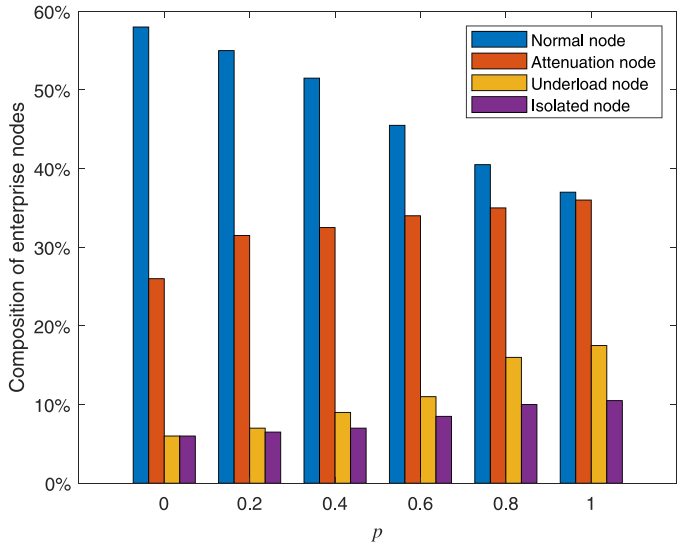


Fig. 19. Composition of enterprise nodes varying Δt ($\alpha = 0.8$, $\theta = 2$, $\beta = 0.2$, $r = 0.5$, $\Delta t = 2$).

underload/attenuation enterprise node can return to its normal state. When the value of $p_i^U(t)$ for an underload enterprise node i at time t or the value of $p_i^A(t)$ for an attenuation enterprise node i at time t exceeds this coefficient, the enterprise can resume normal operation after Δt, based on the accumulation of certain conditions. Therefore, in the automotive manufacturing supply chain network, each enterprise can expedite the resumption of normal operations by appropriately reducing the underload/attenuation state-transition coefficient.

4.5.3. *Impact of isolated state-transition coefficient r*

Fig. 20 displays the steady-state value of relative network efficiency E_{end} for the automotive manufacturing supply chain network with different isolated state-transition coefficients after implementing the recovery strategy. When $\Delta t = 2$ and $r = 0$, E_{end} is 0.79, whereas it decreases to 0.6 when $\Delta t = 2$ and $r = 1$. The decrease in E_{end} with increasing r can be reasonably explained using (16) and (17). As r increases, the probability that the isolated state transition function $R_{j,i}(t)$ has a value greater than 0 at time t diminishes. Consequently, the probability of the isolated failed node i returning to its normal working state decreases. Hence, the value of E_{end} correspondingly decreases.

Fig. 21 presents a comparison of the final composition of enterprise nodes after the network reaches a steady state with different values of r. It is evident that the proportion of failed enterprise nodes positively correlates with the isolated state-transition coefficient

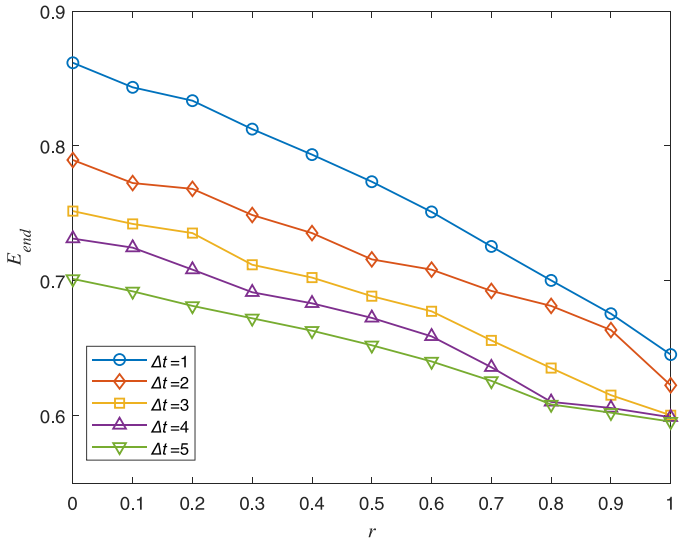


Fig. 20. Relative network efficiency steady-state value E_{end} varying r ($\alpha = 0.8$, $\theta = 2$, $\beta = 0.2$, $p = 0.5$).

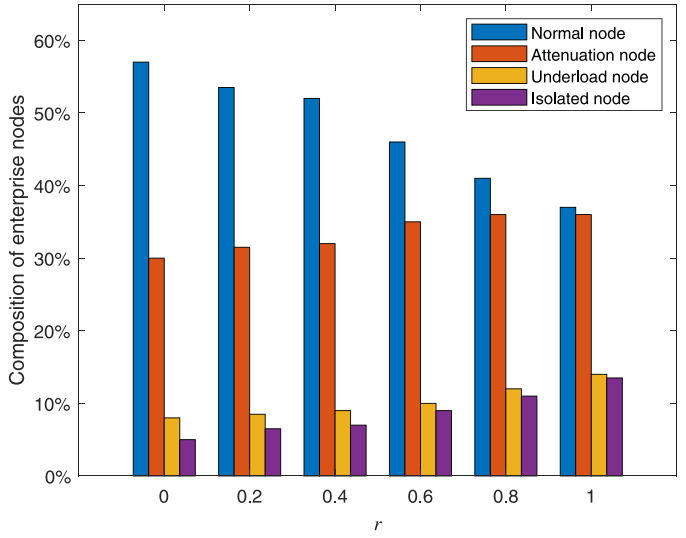


Fig. 21. Composition of enterprise nodes varying r ($\alpha = 0.8$, $\theta = 2$, $\beta = 0.2$, $p = 0.5$, $\Delta t = 2$).

p. Specifically, the number of isolated failed nodes increases more significantly. This is because when the value of $R_{j,i}(t)$ for an isolated enterprise node j at time t exceeds the isolated state-transition coefficient p, the enterprise can establish new business relationships to return to its normal state. The above analysis demonstrates that enterprises can enhance the cascading failure resilience of the network by appropriately relaxing the criteria for establishing business relationships.

4.6. Effectiveness of two-stage recovery strategy

We adopt the HECA strategy to target the top 5% of nodes in the network, and compare the effectiveness of the two-stage recovery strategy with the probabilistic recovery strategy. In [32], the probabilistic recovery strategy is proposed, where failed nodes are randomly recovered after a recovery time interval Δt, based on a probability p_r. After recovery, the nodes maintain their original business relationships. In the two-stage recovery strategy, the adjustable coefficients for recovering failed nodes are the underload/attenuation state-transition coefficient p and the isolated state-transition adjustable coefficient r. Although these coefficients are essentially similar to the failed node recovery probability in the probabilistic recovery strategy, we compare their effects on the relative network efficiency $E(t)$.

The relative network efficiencies $E(t)$ under different recovery strategies are depicted in Figs. 22, 23, and 24. It is evident that the two-stage recovery strategy outperforms the probabilistic recovery strategy irrespective of the values of p, r, and p_r. This can be

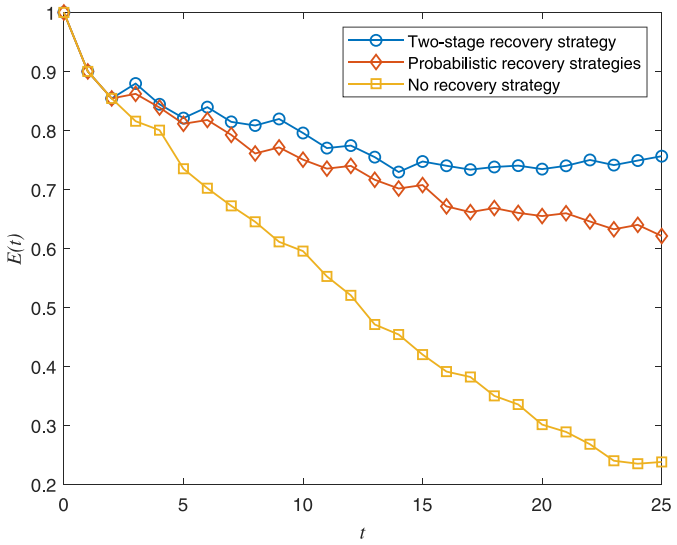


Fig. 22. Relative network efficiency $E(t)$ varying different recovery strategies ($\alpha = 0.6$, $\theta = 2$, $\beta = 0.2$, $\Delta t = 2$, $p = 0.3$, $r = 0.3$, $p_r = 0.3$).

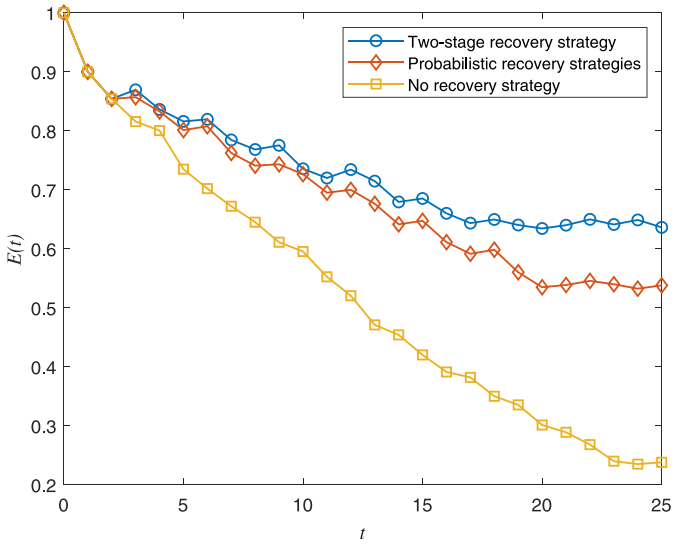


Fig. 23. Relative network efficiency $E(t)$ varying different recovery strategies ($\alpha = 0.6$, $\theta = 2$, $\beta = 0.2$, $\Delta t = 2$, $p = 0.5$, $r = 0.5$, $p_r = 0.5$).

attributed to the fact that the two-stage recovery strategy employs diverse recovery strategies for different failed node components, thereby increasing the probability of successful recovery.

5. Conclusions and the future work

This work develops a realistic cascading failure model to characterize the cascading failure process of automotive manufacturing supply chain networks, considering enterprise roles. Building upon this model, we propose a two-stage recovery strategy aimed at improving the resilience of the network against cascading failures. Extensive experiments are conducted to examine the impact of network modeling parameters, various attack strategies, and recovery-related parameters on the network's cascading failure resilience. Furthermore, the effect of the two-stage recovery strategy has been confirmed by comparing it with other recovery strategies. The aforementioned experimental findings provide several insightful observations:

(1) The primary factor contributing to the decline in cascading failure resilience in the automotive manufacturing supply chain network is the presence of underloaded nodes during load redistribution. To enhance the network's resilience against cascading failures, it is essential to lower the network's capacity lower limit.

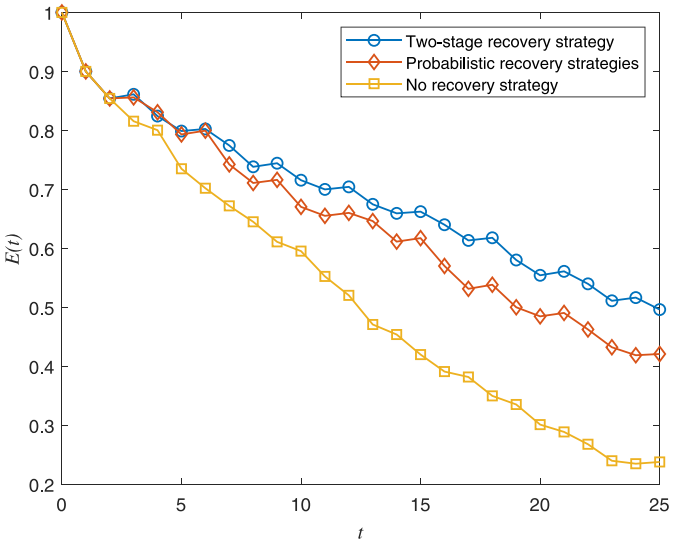


Fig. 24. Relative network efficiency $E_{(t)}$ varying different recovery strategies ($\alpha = 0.6$, $\theta = 2$, $\beta = 0.2$, $\Delta t = 2$, $p = 0.7$, $r = 0.7$, $p_r = 0.7$).

(2) The cascading failure resilience of the automotive manufacturing supply chain network is significantly influenced by the load-exponential coefficient, capacity lower-limit coefficient, and attenuation-tolerance coefficient. These coefficients exhibit a negative correlation with the network's resilience metric. Therefore, it is advisable for enterprise nodes to appropriately reduce these coefficients.
(3) Among the four attack strategies analyzed, the high eigenvector centrality attack strategy and the high degree attack strategy have the most substantial impact on the network's cascading failure resilience. Consequently, protective measures should be developed specifically for high-importance and high-adaptability enterprises within the automotive manufacturing supply chain network.
(4) The longer an enterprise takes to recover, the lower the resilience of the automotive manufacturing supply chain network against damages. Thus, it is crucial to improve the recovery speed of enterprises within the network.
(5) The resilience of the automotive manufacturing supply chain network is positively correlated with the resilience of underload enterprises and their ability to establish business relationships when isolated. Therefore, enterprise nodes should implement appropriate measures to enhance the resilience and the ability of enterprises to build business relationships.
(6) The adoption of a two-stage recovery strategy proves to be more effective in enhancing the cascading failure resilience of the automotive manufacturing supply chain network compared to the commonly used probabilistic recovery strategy. This suggests that formulating a two-stage recovery strategy based on different node states can effectively enhance the network's cascading failure resilience.

In this study, we primarily focused on the cascading failure modeling of automotive manufacturing supply chain networks and the network recovery after cascading failures occur. Many studies [33–35] have indicated that network topology is one of the key factors influencing cascading failures in networks. In our next research work, we will specifically investigate the relationship between cascading failure resilience of the automotive manufacturing supply chain network and its intrinsic topological attributes (e.g., community and cluster). We will also explore how to enhance the network's ability to withstand cascading failures from a topological optimization perspective.

CRediT authorship contribution statement

Xiuwen Fu: Conceptualization, Data curation, Formal analysis, Funding acquisition, Resources, Software, Visualization, Writing – original draft, Writing – review & editing. **Xiaojie Xu:** Conceptualization, Visualization, Writing – original draft, Writing – review & editing. **Wenfeng Li:** Funding acquisition, Supervision.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Xiuwen Fu reports was provided by National Natural Science Foundation of China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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